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1. Let A and B be two sets with $A = \{x, y\}$ and $B = \{a, b\}$. Which one of the following sets represents the cartesian product $A \times B$? 1 / 1 point
- ☒ $\{(x, a), (x, b), (y, a), (y, b)\}$

☐ $\{(x, a), (x, b), (y, a), (y, b), (a, x), (a, y), (b, x), (b, y)\}$

☐ $\{(x, y), (a, b)\}$

☐ $\{(a, x), (b, x), (a, y), (b, y)\}$

✔ Nice work

$A \times B = \{(n, m) : n \in A \text{ and } m \in B\} = \{(x, a), (x, b), (y, a), (y, b)\}$

2. Let A be a consisting of n elements and B be a set containing m . What is the cardinality of $A \times B$? 1 / 1 point
- ☐ m^n

☐ n^m

☒ $n \times m$

☐ $n(m - 1)$

✔ Nice work

$|A \times B| = |A| \times |B| = n \times m$

3. Let A be a set with $A = \{a, b\}$ and let $\mathcal{R}$ be a binary relation on A defined by $\mathcal{R} = \{(a, a), (a, b)\}$. Which one of the following is true? 1 / 1 point
- ☐ $a\mathcal{R}a$ and $b\mathcal{R}a$

☐ $a\mathcal{R}a$ and $b\mathcal{R}b$

☒ $a\mathcal{R}a$ and $a\mathcal{R}b$

☐ $b\mathcal{R}b$ and $b\mathcal{R}a$

✔ Nice work

(a, a) , and $(a, b) \in \mathcal{R}$, hence, $a\mathcal{R}a$ and $a\mathcal{R}b$.

4. Let $\mathcal{R}$ be a binary relation on the set of real numbers, $\mathbb{R}$, defined as by $\mathcal{R} = \{(a, b) \in \mathbb{R}^2 \mid a < b\}$. Which one of the following is true? 1 / 1 point
- ☐ $x\mathcal{R}2x, \forall x \in \mathbb{R}$

☒ $1\mathcal{R}2$

☐ $x\mathcal{R}x, \forall x \in \mathbb{R}$

☐ $2\mathcal{R}1$

✔ Nice work

$1 < 2$, hence, $1\mathcal{R}2$